HIKET — Bayesian Calibration

Methods and pipeline documentation

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Contents

1 Overview

This document describes the Bayesian calibration pipeline for the HIKET multi-model SOC intercomparison study. The calibration is implemented in a model-agnostic engine (`calibration_engine.R`) and four model-specific orchestration scripts (`run_{Model}_calibration.R`).

Models in the intercomparison: Yasso07, Yasso15, Yasso20, RothC (pending).

Core research question: Do divergent model projections of Finnish forest carbon sink saturation reflect genuine structural differences between models, or calibration artefacts? All models are calibrated against the same data with the same statistical framework; structural differences are designed to appear in residuals rather than being absorbed into error terms.

Pipeline split:

Script	Purpose
`Data_work.R`	Data preparation
`run_{Model}_calibration.R`	MCMC calibration, posterior samples
`run_{Model}_predictive.R`	Posterior predictive trajectories, obs vs pred
`run_residual_analysis.R`	Random forest residual analysis (model-agnostic)

2 Models

2.1 Yasso07

The Yasso07 model (Tuomi et al. 2009, 2011) describes SOC decomposition as first-order kinetics in five pools: A (acid-soluble), W (water-soluble), E (ethanol-soluble), N (non-soluble/lignin), and H (humus). Litter enters the A, W, E, N pools proportionally to AWEN fractions; material flows between AWEN pools (controlled by transfer fractions p_{ij}) and to H; H decomposes irreversibly.

The climate modifier is applied multiplicatively to all decomposition rates:

$$\xi(T, T_{\text{amp}}, P) = \exp(\beta_1 T + \beta_2 T^2) \times (1 - \exp(\gamma\, P/1000))$$

where T is mean temperature (°C), P is annual precipitation (mm), and β_1, β_2, γ are free parameters. A woody decomposition size modifier scales decomposition rates for FWL/CWL:

$$h_S(d) = \min\{1,\ (1 + \delta_1 d + \delta_2 d^2)^{-|r|}\}$$

where d is woody litter diameter (cm). Model core implemented in Fortran; called via `.Fortran()`.

2.2 Yasso15 / Yasso20

Structurally identical AWEN pool structure as Yasso07, with pool-group-specific climate responses: separate $(\beta_1, \beta_2, \gamma)$ triplets for the AWE pool group, N pool, and H pool. Both Yasso15 and Yasso20 share the same Fortran core and the same 35-parameter vector structure. The sole computational difference is the temperature integration method: Yasso15 uses a four-point Gaussian approximation of the seasonal temperature cycle, while Yasso20 averages $\exp(\beta_1 T + \beta_2 T^2)$ over all 12 monthly temperatures directly.

Yasso15 includes leaching parameters $w_1, \dots, w_5$ (positions 17–21 of the parameter vector) representing dissolved organic carbon export. In Yasso20, these positions are retained for structural compatibility but are explicitly ignored by the Fortran routine (`mod5c20`, confirmed from source). In Yasso20, leaching is controlled by a fixed scalar argument rather than calibrated parameters. The original Yasso20 calibration (FMI GitHub repository, https://github.com/YASSOmodel) fixed w_1–w_5 at zero; they are retained in `FIXED_RATE_NAMES` accordingly.

Calibration follows the same architecture as Yasso07 with an expanded parameter vector.

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3 Parameter Specification — Yasso07

3.1 Fixed Parameters

Six parameters are fixed at published values (Tuomi et al. 2009, 2011) and are not calibrated:

HIKET name	Published notation	Physical meaning
`alpha_A`, `alpha_W`, `alpha_E`, `alpha_N`	$\alpha_A, \alpha_W, \alpha_E, \alpha_N$	Pool decomposition rate constants (yr^{-1})

HIKET name	Published notation	Physical meaning
`p_H`	p_H	Fraction of decomposed mass entering humus
`alpha_H`	α_H	Humus decomposition rate (yr^{-1})

Rationale for fixing: Rate constants are well-constrained across the international Yasso literature. Calibrating them against Finnish SOC alone risks overfitting to Finnish climate and litter inputs. Structural differences between models in this intercomparison are encoded in flow fractions and climate response — the scientifically relevant comparison axes.

3.2 Free Parameters

20 parameters are jointly calibrated (18 physical + 2 auxiliary uncertainty):

Inter-pool flow fractions (12): p_{AW}, p_{AE}, p_{AN}, p_{WA}, p_{WE}, p_{WN}, p_{EA}, p_{EW}, p_{EN}, p_{NA}, p_{NW}, p_{NE}. Each donor pool i must satisfy $\sum_j p_{ij} \leq B = 1 - p_H$ (mass balance with humus formation).

Climate response (3): β_1 (temperature, linear; log-transformed), β_2 (temperature, quadratic; unconstrained), γ (precipitation; unconstrained).

Woody litter size modifier (3): δ_1 (linear diameter effect; unconstrained), δ_2 (quadratic diameter effect; **log-transformed**), r (power; **log-transformed**).

Auxiliary uncertainty parameters (2):

- σ_{init}: multiplicative initial-condition uncertainty. Added in quadrature to σ_{obs} at the first observation per plot. Captures error from assuming the 1985 SOC equals the analytical steady state (log-transformed; strictly positive).
- σ_{input}: global litter scaling multiplier. Applied to all litter inputs before model evaluation (steady state and transient run consistently). Captures allometric uncertainty in NFI litter estimates ($ $20–50% intrinsic uncertainty). Global (same multiplier for all plots, species, and years). Log-transformed; centred at 1 (no scaling).

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4 Parameter Specification — Yasso15 / Yasso20

4.1 Fixed Parameters

Same six rate constants as Yasso07 ($\alpha_A, \alpha_W, \alpha_E, \alpha_N, p_H, \alpha_H$), fixed at the Yasso15/20 published values (Tuomi et al. 2011 and repository; see Section 7).

Yasso15 and Yasso20: leaching parameters w_1–w_5. Positions 17–21 of the parameter vector are explicitly marked "IGNORED IN THIS FUNCTION'' in both the Yasso15 and Yasso20 Fortran sources. Leaching enters the model via an external scalar argument (`leac`); the HIKET R wrappers set this to zero, disabling leaching in both models. The w parameters are therefore inert in the HIKET calibration and are included in `FIXED_RATE_NAMES`. How the original FMI calibration used these parameters is unknown, as no R wrapper is provided in the Yasso15 repository; the non-zero

w values in the published Yasso15 posterior may reflect a different calling convention or simply unidentified parameters.

Yasso20 additional fixed parameters: six transfer fractions are structurally fixed at zero in the original Yasso20 calibration (posterior SD ≈ 0 across all 10,000 repository samples). These are fixed in the HIKET calibration and not sampled:

$$p_{NW} = p_{AE} = p_{WE} = p_{NE} = p_{AN} = p_{EN} = 0$$

4.2 Free Parameters

Yasso15: 24 physical + 2 auxiliary uncertainty = 26 free parameters. **Yasso20:** 13 physical + 2 auxiliary uncertainty = 15 free parameters (structural zeros removed).

Inter-pool flow fractions: same 12 as Yasso07 (Yasso15); subset for Yasso20.

Pool-group-specific climate response (9 for Yasso15; 9 for Yasso20):

HIKET name	Physical meaning
`beta1`, `beta2`, `gamma`	Temperature and precipitation response for AWE pools
`betaN1`, `betaN2`, `gammaN`	Temperature and precipitation response for N pool
`betaH1`, `betaH2`, `gammaH`	Temperature and precipitation response for H pool

Woody litter size modifier (3): δ_1, δ_2, r as in Yasso07.

Leaching parameters (5, Yasso15 only): $w_1, \ldots, w_5$. Represent dissolved organic carbon export from each pool group. All fixed at zero in Yasso20.

Auxiliary uncertainty parameters (2): σ_{init}, σ_{input} as in Yasso07.

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5 Parameter Naming Conventions

The three model versions use different parameter naming conventions across publications, original calibration files, and this HIKET implementation. The table below reconciles all conventions for the non-fraction parameters (fractions follow a consistent p_{XY} notation across all sources).

HIKET code name	`y07par_gui.csv`	`y15par.csv`	Fortran `theta(i)` (Y15/20)	Paper notation	Physical meaning
`alpha_A`	`aA`	`aA`	θ_1	α_A	A pool rate (yr^{-1})
`alpha_W`	`aW`	`aW`	θ_2	α_W	W pool rate (yr^{-1})
`alpha_E`	`aE`	`aE`	θ_3	α_E	E pool rate (yr^{-1})

HIKET code name	y07par_gui.csv	y15par.csv	Fortran theta(i) (Y15/20)	Paper notation	Physical meaning
alpha_N	aN	aN	θ_4	α_N	N pool rate (yr^{-1})
beta1	b1	b1	θ_{22}	β_1	Temp. linear (AWE)
beta2	b2	b2	θ_{23}	β_2	Temp. quadratic (AWE)
gamma	gamma	g	θ_{28}	γ	Precip. (AWE)
betaN1	—	bN1	θ_{24}	β_{N1}	Temp. linear (N)
betaN2	—	bN2	θ_{25}	β_{N2}	Temp. quadratic (N)
gammaN	—	gN	θ_{29}	γ_N	Precip. (N)
betaH1	—	bH1	θ_{26}	β_{H1}	Temp. linear (H)
betaH2	—	bH2	θ_{27}	β_{H2}	Temp. quadratic (H)
gammaH	—	gH	θ_{30}	γ_H	Precip. (H)
delta1	phi1	th1	θ_{33}	ϕ_1	Size modifier linear
delta2	phi2	th2	θ_{34}	ϕ_2	Size modifier quadratic
r	r	r	θ_{35}	r	Size modifier power
alpha_H	aH	aH	θ_{32}	α_H	Humus rate (yr^{-1})
p_H	pH	pH	θ_{31}	p_H	Humus fraction
w1–w5	—	w1–w5	θ_{17}–θ_{21}	w_1–w_5	Leaching (Yasso15); ignored in Yasso20

Fortran index reference. The Fortran theta(i) column above refers to the original Yasso15/20 Fortran subroutine (mod5c20) from the FMI GitHub repository (https://github.com/YASSOmodel), not the HIKET reimplementation. The HIKET wrapper reorders parameters internally via ORIG_TO_LORENZO before calling the Fortran.

Critical ordering note. The original Yasso15/20 Fortran and .dat files use the ordering $(\alpha, p_{ij}, w_{1:5}, \beta, \beta_N, \beta_H, \gamma, \gamma_N, \gamma_H, p_H, \alpha_H, \delta_1, \delta_2, r)$, placing leaching before climate parameters. The HIKET implementation reorders to $(\alpha, p_{ij}, \beta, \gamma, \beta_N, \gamma_N, \beta_H, \gamma_H, p_H, \alpha_H, \delta_1, \delta_2, r, w_{1:5})$. The bijective column remapping is documented in Priors_model_matching.R (vector ORIG_TO_LORENZO).

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6 Parameter Transformations

The DEzs sampler operates in unconstrained $\mathbb{R}^n$ where $n = 20$ (Yasso07), 26 (Yasso15), or 15 (Yasso20). Physical parameters are recovered by applying to_original() to every proposal.

6.1 Stick-Breaking (flow fractions)

Each of the four donor columns (A, W, E, N) has three outflows satisfying:

$$p_{ij} \geq 0 \quad \forall j, \qquad \sum_j p_{ij} \leq B$$

The stick-breaking transform converts a vector $v \in \mathbb{R}^3$ to a valid fraction vector p:

$$p_i = \sigma(v_i) \times r_i, \qquad r_{i+1} = r_i - p_i, \quad r_1 = B$$

where $\sigma(v) = (1 + e^{-v})^{-1}$ is the logistic sigmoid. The inverse (physical $\rightarrow$ unconstrained) follows by sequential logit transformation of the normalised remainder fractions.

6.2 Log Transform

Applied to parameters that must be strictly positive:

$$v = \log(\theta), \qquad \theta = e^v$$

Applied to: σ_{init}, σ_{input}, β_1, β_{N1}, β_{H1} (temperature sensitivities, must be positive for boreal decomposition; see Section 7), δ_2 (size modifier quadratic, must be positive to prevent numerical failure; see below), r (size modifier power; only $|r|$ is used in Fortran).

6.3 Unconstrained

No transformation: β_2, β_{N2}, β_{H2}, γ, γ_N, γ_H, δ_1, $w_{1:5}$.

6.4 The $\delta_2 > 0$ Requirement

The size modifier is $h_S(d) = \min\{1, (1 + \delta_1 d + \delta_2 d^2)^{-|r|}\}$. The expression inside the brackets must stay positive for all realistic woody litter diameters d (Finnish stumps: 2–30 cm). With $\delta_2 > 0$ the quadratic opens upward and stays positive for all $d \geq 0$ given the published values ($\delta_1 = -1.71$). With $\delta_2 < 0$ the quadratic opens downward, reaches zero at some diameter d_0, and the expression becomes negative beyond that point. Raising a negative number to the non-integer power $-|r|$ is undefined in Fortran and propagates as NaN, which returns $-\infty$ to the likelihood.

This is not a HIKET-specific issue: the same formula appears verbatim in both the original Yasso15 and Yasso20 Fortran sources from the FMI repository. Any parameter set with $\delta_2 < 0$ would produce NaN in the original code when applied to woody litter larger than a few millimetres. The problem did not manifest in the original calibrations because those datasets apparently contained little variation in woody litter diameter, leaving δ_2 unidentified and allowing it to drift negative without penalty.

Applying a log transform to δ_2 in the HIKET calibration enforces $\delta_2 > 0$ throughout the MCMC. The published defaults ($\delta_2 = +0.86$ for Yasso07, $+1.27$ for Yasso15) confirm the physically correct sign.

The log transform on r is a minor complement: since only $-|r|$ enters the Fortran expression, the sign of r is irrelevant. Sampling in unconstrained space therefore wasted half the prior support on numerically identical values.

6.5 Jacobian Correction

When parameters are reparameterised (e.g. the sampler proposes $v = \log \delta_2$ instead of δ_2 itself), the probability density must be corrected for the stretching or compression introduced by the transformation. Intuitively: if physical space near $\delta_2 = 0.1$ is much more "squeezed'' than near $\delta_2 = 10$ after the log transform, proposals that land near zero are more probable in log space than they should be in physical space. The Jacobian is the correction factor that compensates for this distortion, ensuring the prior and posterior remain correctly normalised regardless of the parameterisation chosen for the sampler.

Formally, the log-Jacobian $\log |\partial\theta/\partial v|$ is added to the log-likelihood at every evaluation:

$$\log p(\theta \mid y) \propto \log p(y \mid \theta) + \log p(\theta) + \log \left| \frac{\partial \theta}{\partial v} \right|$$

For stick-breaking, the Jacobian factor per element is $\sigma(v_i)(1-\sigma(v_i)) \times r_i$. For the log transform it is $e^v = \theta$. The Jacobian is added to the likelihood (not the prior) so that the BayesianTools prior object remains a standard Gaussian.

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7 Prior Distribution

7.1 Philosophy

The prior is a multivariate normal in unconstrained space, independent across parameters:

$$p(v) = \prod_k \mathcal{N}(v_k;\ v_k^*,\ \sigma_k^2)$$

Three parameter groups receive qualitatively different prior treatment:

1. **Climate and size modifier parameters:** informative priors derived from the original global calibration datasets (Tuomi et al. 2008–2011) or from the published Yasso15/20 MCMC posteriors (FMI GitHub repository, https://github.com/YASSOmodel). These parameters are well-constrained by external data; tight priors prevent degenerate compensation between parameters (particularly the γ–σ_{input} loop identified during initial runs) and prevent exploration of physically invalid regions.

2. **Transfer fractions:** weakly informative. The prior width in unconstrained stick-breaking space is set to 1.0 for all fractions. The stick-breaking transform already guarantees that fractions are non-negative and sum to at most $B = 1 - p_H$ regardless of the prior width; the width of 1.0 simply means the sampler explores this constrained space broadly without strong directional preference. Finnish data should freely inform the pool flow structure — this is the primary structural comparison axis between models in the HIKET intercomparison.

3. **Auxiliary uncertainty parameters (σ_{init}, σ_{input}):** moderately informative, with prior width 0.5 in log space. σ_{init} is the proportional standard deviation of the initial-condition error: $\sigma_{\text{init}} = 0.10$ means the 1985 predicted SOC is uncertain by roughly $\pm 10\%$ (one sigma), reflecting that real forests are not perfectly at steady state. A prior width of 0.5 in log space gives a physical 95% CI of approximately $[0.04, 0.27]$, i.e.
between 4% and 27% initial uncertainty. σ_{input} is a global multiplicative scaling on all litter inputs (see Section on per-plot likelihood evaluation). Prior width 0.5 in log space gives physical 95% CI $\approx [0.37, 2.72]$, wide enough to accommodate systematic allometric uncertainty in NFI litter estimates.

7.2 Prior Centres

Yasso07: prior centres are the published parameter values from Tuomi et al. (2009, 2011), confirmed to agree with `YASSO07_DEFAULT_PARAMS` in the wrapper to within numerical precision. Decomposition rates in `y07par_gui.csv` are stored as positive absolute values; the wrapper correctly uses them with a negative sign in the transfer matrix.

Yasso15: prior centres for climate and size modifier parameters are set to the posterior means from the original Yasso15 global calibration (FMI repository, `Yasso15.dat`, 10,000 thinned MCMC samples after dropping the MAP row). Transfer fractions use the Yasso15 model defaults. The posterior means agree with the published MAP values in `y15par.csv` to four significant figures after applying the column remapping described in Section **??**.

Yasso20: prior centres for climate and size modifier parameters are set to the posterior means from the original Yasso20 global calibration (FMI repository, `Yasso20.dat`, 10,000 samples). Transfer fractions use Yasso20 model defaults.

Auxiliary uncertainty parameters: σ_{init} centred at 0.10 (10% initial condition uncertainty), σ_{input} centred at 1.00 (no a priori scaling adjustment).

7.3 Prior Widths — Yasso07

No MCMC posterior samples are available from the Yasso07 original calibration. Prior widths are set from two sources: (a) published 95% credible intervals from Tuomi et al. 2010 Table 4 for the size modifier parameters; (b) physical range arguments for the climate parameters, based on what range of climate modifiers is plausible at Finnish temperatures and precipitation.

Parameter	Prior width	Space	Basis
Flow fractions	1.0	unconstrained	Weakly informative; Finnish data informs structure
β_1	0.20	log	Physical range: $\exp(0.2 \times 10) \approx 7\times$ modifier at $T = 10°\text{C}$ gives plausible boreal range

Parameter	Prior width	Space	Basis
β_2	0.05	physical	Physical range: prevents modifier curves with implausible curvature
γ	0.30	physical	Physical range: keeps precip. modifier in $[0.3, 0.6]$ at Finnish $P = 500$–650 mm
δ_1	0.15	physical	Tuomi et al. 2010 Table 4: published 95% CI ± 0.16 cm^{-1}
δ_2	0.10	log	Tuomi et al. 2010 Table 4: published 95% CI ± 0.10 cm^{-2}
r	0.015	log	Tuomi et al. 2010 Table 4: published 95% CI ± 0.013
σ_{init}	0.5	log	Physical 95% CI $\approx [0.04, 0.27]$: reasonable range for steady-state error
σ_{input}	0.5	log	Physical 95% CI $\approx [0.37, 2.72]$: covers NFI allometric uncertainty

7.4 Prior Widths — Yasso15 and Yasso20

For Yasso15/20, prior widths for climate and size modifier parameters are derived from the standard deviations of the published MCMC posterior samples from the FMI GitHub repository (https://github.com/YASSOmodel, files `Yasso15.dat` and `Yasso20.dat`, 10,000 thinned samples each). For log-transformed parameters (β_1, β_{N1}, β_{H1}, δ_2, r) the width is SD($\log\theta$) across posterior samples; for unconstrained parameters it is SD(θ).

The γ_H (H pool precipitation sensitivity) prior width is capped at 1.5 in both models. At Finnish precipitation ($P \approx 600$ mm), $1 - \exp(\gamma_H \times 0.6) \approx 1$ for any $\gamma_H < -5$, meaning small changes in γ_H produce negligible changes in the H pool climate modifier. The large posterior SD for γ_H (3.99 for Yasso15, 1.39 for Yasso20) is consistent with this near-unidentifiability in the original global calibration dataset; the cap of 1.5 uses the better-constrained Yasso20 posterior as a reference. Note that the Yasso development team is currently reviewing whether pool-specific precipitation sensitivity is warranted; a common γ across pools may be adopted in future versions.

Note on terminology: throughout this document, **free** parameters are those sampled by the MCMC, as opposed to **fixed** parameters held at published values before calibration begins.

Parameter	Yasso15 σ_k	Yasso20 σ_k	Space	Note
Flow fractions	1.0	1.0	unconstrained	Weakly informative
β_1	0.046	0.104	log	SD(log β_1) from posterior
β_2	0.00014	0.00054	physical	Very tight; well-identified
γ	0.070	0.146	physical	SD(γ) from posterior
β_{N1}	0.105	0.062	log	SD(log β_{N1})
β_{N2}	0.00008	0.00041	physical	
γ_N	0.155	0.035	physical	
β_{H1}	0.135	0.091	log	SD(log β_{H1})
β_{H2}	0.00016	0.00009	physical	
γ_H	**1.50**	1.391	physical	Capped (identifiability; see above)
δ_1	0.328	0.435	physical	SD(δ_1)
δ_2	0.286	0.211	log	SD(log δ_2)
r	0.055	0.054	log	SD(log r)
σ_{init}	0.5	0.5	log	Fixed
σ_{input}	0.5	0.5	log	Fixed

7.5 Physical Sign Constraints

Several parameters must satisfy sign constraints from physical or mathematical requirements. These are enforced via log transforms rather than prior bounds, which would introduce hard edges incompatible with the Gaussian prior assumption.

Constraint	Parameters	Basis
> 0	$\beta_1, \beta_{N1}, \beta_{H1}$	Decomposition increases with temperature in boreal Finland (-5 to $+20°$C). Enforced by log transform.
> 0	δ_2	Size modifier $h_S(d) > 0$ for all Finnish diameters requires positive quadratic. Negative δ_2 causes NaN in Fortran for $d \gtrsim 0.4$ cm. Enforced by log transform.
> 0	r	Only $-\lvert r\rvert$ enters Fortran. Sign is irrelevant; log transform avoids wasting half the prior on sign-equivalent values.
< 0	$\gamma, \gamma_N, \gamma_H$	$1 - \exp(\gamma P/1000) > 0$ requires $\gamma < 0$. Enforced by tight prior centred on negative published values; all posteriors respect this sign.

Note on γ_H: enforcing $\gamma_H < 0$ via $\log(-\gamma_H)$ transform would require negating γ_H in `assemble_model_params()`. The current implementation retains γ_H as unconstrained with a tight prior centre (Yasso15: -12.54; Yasso20: -7.63) and a capped sigma; the likelihood itself strongly enforces the negative sign.

7.6 Published Parameter Values at Prior Centres

The tables below give the physical-space prior centres for the climate and size modifier parameters (those sampled by the MCMC; see the note on *free* vs *fixed* in Section 7.2). Transfer fraction prior centres are the respective model's published defaults and are not listed separately.

Yasso07 (from Tuomi et al. 2009, 2011; `y07par_gui.csv`):

Parameter	Value	Source
β_1	0.09873	Tuomi et al. 2009
β_2	$-$0.00157	Tuomi et al. 2009
γ	$-$1.2717	Tuomi et al. 2009
δ_1	$-$1.7084	Tuomi et al. 2010
δ_2	$+$0.8586	Tuomi et al. 2010
r	$-$0.3068	Tuomi et al. 2010

Yasso15 (posterior means from FMI repository `Yasso15.dat`; agree with `y15par.csv` MAP to four significant figures):

Parameter	Value
β_1	0.0906
β_2	$-$0.000215
γ	$-$1.8090
β_{N1}	0.04878
β_{N2}	$-$0.0000792
γ_N	$-$1.1729
β_{H1}	0.03518
β_{H2}	$-$0.000208
γ_H	$-$12.541
δ_1	$-$0.4388
δ_2	$+$1.2684
r	$+$0.2569

Yasso20 (posterior means from FMI repository `Yasso20.dat`):

Parameter	Value
β_1	0.1666
β_2	$-$0.00236
γ	$-$1.5641
β_{N1}	0.17959

Parameter	Value
β_{N2}	$-$0.00510
γ_N	$-$1.9653
β_{H1}	0.06987
β_{H2}	$-$0.0000980
γ_H	$-$7.6255
δ_1	$-$2.4287
δ_2	$+$1.1270
r	$+$0.2539

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8 Likelihood

8.1 Observation Model

A multiplicative-normal (proportional-error) model is used:

$$\mathrm{SOC}_{\mathrm{obs},t} \sim \mathcal{N}(\hat{C}_t,\ s_t^2)$$

where the standard deviation scales with the predicted SOC:

$$s_t = \begin{cases} \hat{C}_t \sqrt{\sigma_{\mathrm{obs}}^2 + \sigma_{\mathrm{init}}^2} & \text{first observation per plot (1985)} \\ \hat{C}_t\, \sigma_{\mathrm{obs}} & \text{subsequent observations (2006, 2024)} \end{cases}$$

Rationale for multiplicative-normal: A log-normal error model was used during a period when the data contained a $\times 10$ litter overestimate (since corrected). With litter inputs in the correct boreal range, a multiplicative-normal model is appropriate.

8.2 Fixed Observation Error

σ_{obs} is **fixed** at the observed coefficient of variation of SOC across all calibration plots:

$$\sigma_{\mathrm{obs}} = \frac{\mathrm{sd}(\mathrm{SOC}_{\mathrm{obs}})}{\mathrm{mean}(\mathrm{SOC}_{\mathrm{obs}})} \approx 0.45$$

This value is **identical across all four models**. Fixing σ_{obs} ensures structural differences appear in residuals rather than being absorbed into the error term.

8.3 Log-Likelihood

For each plot i with n_i observations:

$$\ell_i(\theta) = \sum_{k=1}^{n_i} \log \mathcal{N}(\mathrm{SOC}_{i,k};\ \hat{C}_{i,t_{ik}},\ s_{i,k}^2)$$

Total log-posterior in unconstrained space:

$$\mathcal{L}(v) = \sum_i \ell_i(\text{to\_original}(v)) + \log\left|\frac{\partial\theta}{\partial v}\right|$$

If any plot returns $-\infty$, the entire proposal is rejected. The inner per-plot function is byte-compiled with `cmpfun()` for a ~10–20% speedup at the R level.

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9 Per-Plot Likelihood Evaluation

Each likelihood evaluation executes three stages per plot in parallel via `mclapply`.

9.1 Stage 1: Environmental Modifier Time Series

For each year t in 1985–2023, `compute_xi()` computes the scalar modifier ξ_t from annual climate summaries (mean temperature, temperature amplitude, annual precipitation). The modifier is applied multiplicatively to all decomposition rates.

9.2 Stage 2: Steady-State Initialisation

The initial SOC pool distribution C^* (5-element vector) is computed analytically:

$$C^* = -A(\theta,\ \xi_{\text{ss}})^{-1}\, b(\theta)$$

where A is the transfer matrix, b is the mean litter input vector, and ξ_{ss} is the climate modifier evaluated at mean climate over the **20-year steady-state window** (1985–2004).

Jensen's inequality correction. Because ξ is nonlinear in T and P:

$$\xi(\bar{T}, \bar{P}) \neq \overline{\xi(T_t, P_t)}$$

Using the average of annual modifiers instead of the modifier at average climate introduces a 5–15% systematic bias in boreal conditions. The correct approach is enforced through the `compute_xi_mean()` interface: it averages T, T_{amp}, and P over the steady-state window and calls `compute_xi()` once.

Litter scaling. σ_{input} is applied to litter inputs in both the steady-state initialisation and the transient run, so the calibrated multiplier affects the initial condition and the trajectory consistently.

9.3 Stage 3: Transient Simulation

Starting from C^*, the model runs forward 1985–2023 using the annual ξ_t time series and actual annual litter inputs. A single Fortran call per plot returns the full pool trajectory; predicted SOC at observation years (1985, 2006, 2024) is extracted and compared to observations.

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10 MCMC Sampling

10.1 Sampler

DEzs (Differential Evolution with snooker updates; ter Braak & Vrugt 2008), implemented in `BayesianTools` (Hartig et al. 2023). Selected for:

- No gradient requirements (black-box Fortran model).
- Robust performance on correlated, moderately high-dimensional posteriors (15–26 free parameters).
- Population-based proposals that efficiently explore posterior ridges from compositional constraints.

10.2 Settings

Setting	Test (Mac)	Production (Puhti)
Chains	4	4
Iterations	10,000	50,000
Burn-in	2,000	10,000
DEzs jitter ε	0.001	0.001
Random seed	2025	2025

Burn-in fraction: 20%. Four chains give meaningfully tighter Gelman–Rubin bounds than three chains for negligible extra cost.

10.3 Parallelisation

Sequential-chains (current): chains run one after another (`lapply`), each using all available cores for the `mclapply` per-plot loop. On Puhti: 39 cores per chain (`parallelly::availableCores()`). On Mac: `detectCores() - 1`.

Hybrid (future Puhti production): outer `lapply` replaced by `mclapply(mc.cores = 4)` with `CORES_PER_CHAIN = 90`, giving $4 \times 90 = 360$ of 384 cores. Nested forking is supported on Linux but not macOS, so sequential is mandatory for Mac development.

`parallel = FALSE` is passed to `createBayesianSetup()` so BayesianTools' built-in parallelism is suppressed.

10.4 Starting Point

All chains start from `best_x = to_unconstrained(published_defaults)`. DEzs initialises its population from small random perturbations around `best_x` and rapidly disperses via differential-evolution proposals.

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11 Pre-MCMC Sanity Check

Before launching MCMC, a forward simulation at the prior centre (published defaults) is run for 4 randomly selected plots:

1. **Numerical stability:** all predicted SOC finite and positive.
2. **Order-of-magnitude check:** predicted / observed $\in [0.1, 10]$ at all observation years.
3. **Likelihood check:** `ll_fn(best_x)` finite across all calibration plots.

Output: `{Model}_forward_at_defaults_{RUN_ID}.png` (2$×$2 panel). If the sanity check fails, MCMC is not launched.

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12 Convergence Diagnostics

All diagnostics computed on post-burnin samples. $\hat{R}$ and ESS computed in unconstrained space; posterior summaries reported in physical space via `to_original()`.

12.1 Gelman–Rubin $\hat{R}$

$\hat{R}$	Interpretation
< 1.05	Good convergence
1.05–1.10	Marginal; extend chains
> 1.10	Poor convergence; run not usable

Computed both per-parameter (univariate, `coda::gelman.diag()`) and jointly (multivariate $\hat{R}$).

12.2 Effective Sample Size

`coda::effectiveSize()`. Values < 200 indicate poor mixing.

12.3 Diagnostic Outputs

All files written to `Calibration_real_data/diagnostics/{Model}/`:

File	Content
`*_traces_*.png`	Trace plots per parameter, one line per chain
`*_marginals_fractions_*.png`	Prior vs posterior KDE for the 12 flow fractions
`*_marginals_climate_aux_*.png`	Prior vs posterior KDE for climate and auxiliary uncertainty parameters
`*_rhat_barplot_*.png`	$\hat{R}$ per parameter, coloured by convergence zone
`*_kl_divergence_*.png`	KL(posterior $\|$ prior) per parameter
`*_report_*.txt`	$\hat{R}$, ESS, posterior quantile table, chain health

Prior and posterior KDEs are evaluated on the same x-grid derived from the prior's support, not the posterior's range, to avoid misleading compression.

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13 Output Files

Written to `Calibration_real_data/runs/`:

File	Content	Used by
`{Model}_chains_{RUN_ID}.rds`	Raw BayesianTools chain objects (unconstrained)	Chain extension, re-diagnostics
`{Model}_posterior_{RUN_ID}.rds`	Post-burnin samples in **physical space**	Predictive script, downstream analysis
`{Model}_metadata_{RUN_ID}.rds`	Run config, R version, wallclock, summary stats	Reproducibility
`{Model}_inputs_{RUN_ID}.rds`	Per-plot structures: `climate_by_plot`, `inputs_by_plot`, `litter_means`, `obs_meta`, `plot_info`	Predictive script

The inputs RDS ensures exact provenance: the predictive script reads the same per-plot structures the calibration used, even if `Data_work.R` is subsequently updated.

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14 Key Design Decisions

Architectural decisions locked in after discussion (April–May 2026):

1. **Multiplicative-normal likelihood.** Reverted from a log-normal workaround added when litter inputs had a $\times 10$ overestimate. With correct litter magnitudes, multiplicative-normal is the appropriate model.

2. **Fixed σ_{obs} across all models.** Ensures structural differences appear in residuals. A poorly fitting model cannot mask its inadequacy by calibrating a larger error variance.

3. **$\xi(\bar{T}, \bar{P})$, not $\overline{\xi(T_t, P_t)}$, for steady state.** Jensen's inequality correction; 5–15% bias in boreal conditions. Fixed at the `compute_xi_mean` interface level in `calibration_engine.R`.

4. **STEADY_STATE_YEARS = 20.** Uses 1985–2004 climate for the steady-state scalar, pre-dating accelerated boreal warming. Sensitivity to this choice is a planned post-paper analysis.

5. **Fixed decomposition rates.** $\alpha_A, \alpha_W, \alpha_E, \alpha_N, p_H, \alpha_H$ fixed at published values. Structural intercomparison should be driven by flow fractions and climate response.

6. **Log transform on δ_2 and r.** Enforces $\delta_2 > 0$ to prevent NaN in the Fortran size modifier and eliminates sign-redundancy in r. Addresses a systematic source of $-\infty$ likelihood evaluations (~24% of proposals in initial unconstrained runs).

7. **Informative priors for climate/size parameters.** Derived from published posteriors (Yasso15/20) or calibration confidence intervals (Yasso07). Prevents the γ–σ_{input} compensation loop observed in initial runs; preserves freedom for Finnish data to inform flow fractions.

8. **Yasso20 structural zeros fixed.** Eleven parameters with SD = 0 in the original Yasso20 calibration are fixed rather than sampled. Sampling near a hard boundary degrades DEzs mixing.

9. **Sequential chains.** Mandatory on macOS. Hybrid strategy (4×39 cores) available on Puhti.

10. **Global σ_{input}.** Single litter scaling multiplier. Per-species stratification is a possible extension but would introduce identifiability questions that distract from the intercomparison.

11. **Three SOC observation years (1985, 2006, 2024).** The Komeetta campaign

(2024) provides a third calibration target. The 2006–2024 stability constrains equilibrium decomposition rates independently of the 1985–2006 accumulation dynamics.

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15 Software

Component	Details
R	$\geq$ 4.3; Puhti: 4.5.2 (`r-env` Apptainer)
`BayesianTools`	Hartig et al. (2023); DEzs sampler
`coda`	Gelman–Rubin $\hat{R}$, ESS
`compiler`	`cmpfun()` byte-compilation
`parallel`	`mclapply()` per-plot parallelism
Fortran model cores	`R CMD SHLIB`; `.Fortran()` interface
HPC	CSC Puhti; `small` partition; 1 node, 40 cores, 2 GB/core

Random seed: `set.seed(2025)` before chain initialisation (consistent across all model scripts). Mac test plot subsampling: `set.seed(42)`.